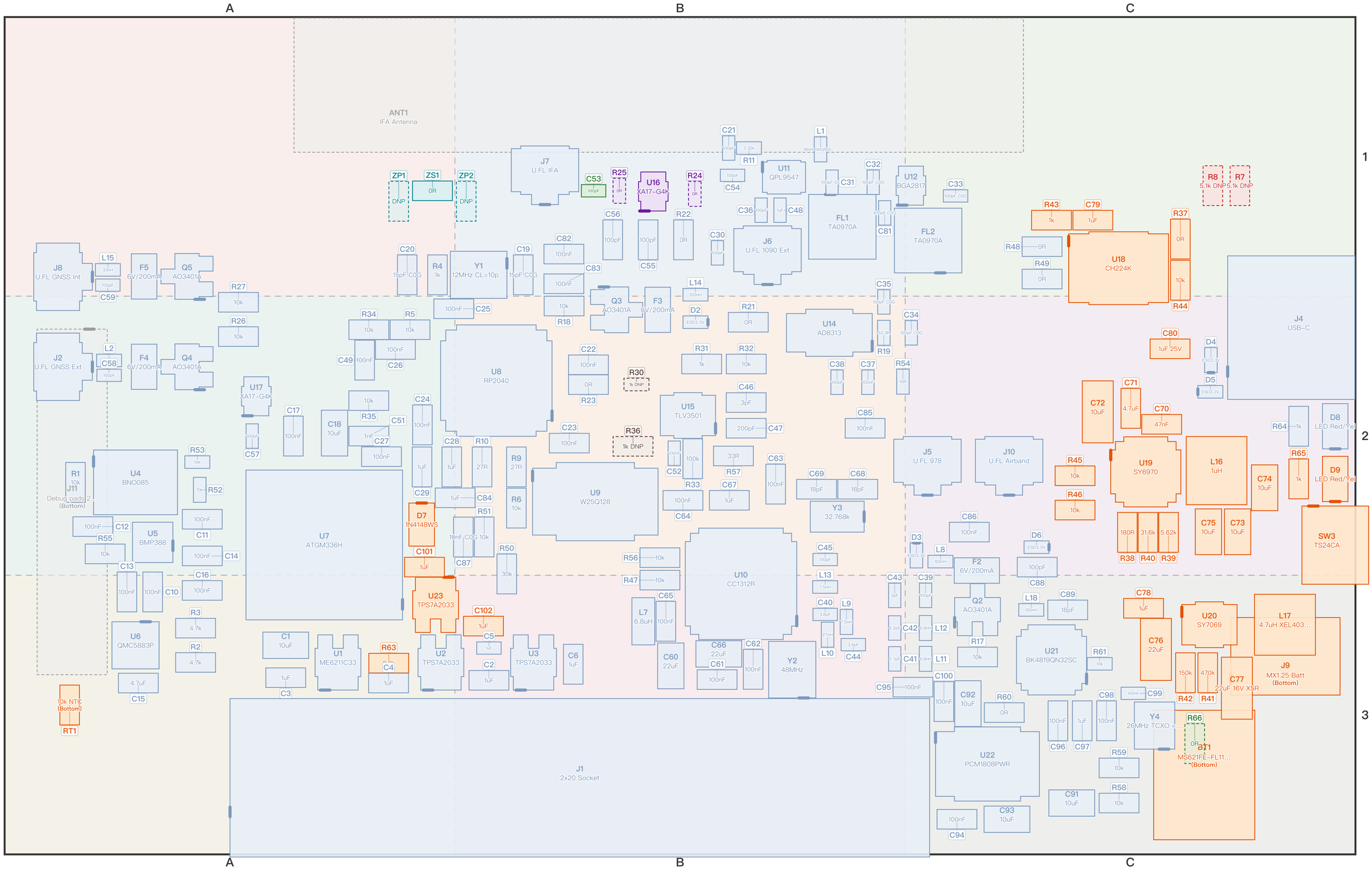
Pilot Kit Expansion Board V4.10 Assembly Map — Powered Variant (Top View)

Parts are drawn to their real outline and orientation; value inside the outline, designator outside. A heavy | marks pin 1 / the polarised end. The nine background tints (columns A/B/C, rows 1/2/3) match the grid badges on page 2. Board 100.0×62.0mm, scaled 5.62×.

Standard 178 | Power Section 37 | No-Power Variant 2 | Antenna Select 3 | IFA Match 3 | Debug Position 2 | DNP 2 | No Part 2 | Dashed = not fitted 11



229 designators (excluding H1~H4 mounting holes and FID1~3 fiducials); 11 of them are not fitted on this variant. | Placement notes on page 3 | Rules: SELECTIVE_PLACEMENT-zh_CN.md

Pilot Kit Expansion Board V4.10 Designator Index — Grid Reference

Each grid badge is the same colour as its cell on page 1 — match the colour to jump straight to it. Flags: ✕ = not fitted, | = polarised (line it up with the polarity mark on page 1). Designators marked ※ have a specific note on page 3.

Ref	Grid	Type	Flags	Value	Footprint
ANT1※	A1	No Part	✕	IFA Antenna	PCB copper
BT1	C3	Power Section		MS621FE-FL11… (Bottom)	MS621 tabs
C1	A3	Standard		10uF	0805
C2	B3	Standard		1uF	0603
C3	A3	Standard		1uF	0603
C4	A3	Standard		1uF	0603
C5	B3	Standard		1uF	0402
C6	B3	Standard		1uF	0603
C10	A3	Standard		100nF	0603
C11	A2	Standard		100nF	0603
C12	A2	Standard		100nF	0603
C13	A3	Standard		100nF	0603
C14	A2	Standard		100nF	0603
C15	A3	Standard		4.7uF	0603
C16	A3	Standard		100nF	0603
C17	A2	Standard		100nF	0603
C18	A2	Standard		10uF	0805
C19	B1	Standard		15pF C0G	0603
C20	A1	Standard		15pF C0G	0603
C21	B1	Standard		100pF	0402
C22	B2	Standard		100nF	0603
C23	B2	Standard		100nF	0603
C24	A2	Standard		100nF	0603
C25	A2	Standard		100nF	0603
C26	A2	Standard		100nF	0603
C27	A2	Standard		100nF	0603
C28	A2	Standard		1uF	0603
C29	A2	Standard		1uF	0603
C30	B1	Standard		100pF	0402
C31	B1	Standard		100pF C0G	0402
C32	B1	Standard		100pF C0G	0402
C33	C1	Standard		100pF C0G	0402
C34	C2	Standard		100pF C0G	0402
C35	B2	Standard		100pF C0G	0402
C36	B1	Standard		100pF	0402
C37	B2	Standard		100nF	0402
C38	B2	Standard		100pF	0402
C39	C3	Standard		100pF	0402
C40	B3	Standard		3.6pF	0402
C41	B3	Standard		2.7pF	0402
C42	B3	Standard		6.2pF	0402
C43	B3	Standard		3pF	0402
C44	B3	Standard		3.6pF	0402
C45	B2	Standard		100pF	0402
C46	B2	Standard		3pF	0603
C47	B2	Standard		200pF	0603
C48	B1	Standard		1uF	0402
C49	A2	Standard		100nF	0603
C51	A2	Standard		1nF	0603
C52	B2	Standard		100nF	0402
C53※	B1	Debug Position		100pF	0402
C54	B1	Standard		100pF	0402
C55	B1	Standard		100pF	0603
C56	B1	Standard		100pF	0603
C57	A2	Standard		100pF	0402
C58	A2	Standard		100pF	0402
C59	A1	Standard		100pF	0402
C60	B3	Standard		22uF	0805
C61	B3	Standard		100nF	0603
C62	B3	Standard		100nF	0603
C63	B2	Standard		100nF	0603
C64	B2	Standard		100nF	0603
C65	B3	Standard		100nF	0603
C66	B3	Standard		22uF	0805
C67	B2	Standard		1uF	0603
C68	B2	Standard		18pF	0603
C69	B2	Standard		18pF	0603
C70	C2	Power Section		47nF	0603
C71	C2	Power Section		4.7uF	0603
C72	C2	Power Section		10uF	1206
C73	C2	Power Section		10uF	0805
C74	C2	Power Section		10uF	0805
C75	C2	Power Section		10uF	0805
C76	C3	Power Section		22uF	1206
C77	C3	Power Section		22uF 16V X5R	1206
C78	C3	Power Section		1uF	0603
C79	C1	Power Section		1uF	0603
C80	C2	Power Section		1uF 25V	0603
C81	B1	Standard		470pF C0G	0402
C82	B1	Standard		100nF	0603
C83	B1	Standard		100nF	0603
C84	B2	Standard		1uF	0603
C85	B2	Standard		100nF	0603
C86	C2	Standard		100nF	0603
C87	B2	Standard		10nF C0G	0603
C88	C2	Standard		100pF	0603
C89	C3	Standard		18pF	0603
C91	C3	Standard		10uF	0805
C92	C3	Standard		10uF	0805
C93	C3	Standard		10uF	0805
C94	C3	Standard		100nF	0603
C95	C3	Standard		100nF	0603
C96	C3	Standard		100nF	0603
C97	C3	Standard		1uF	0603
C98	C3	Standard		100nF	0603
C99	C3	Standard		100nF	0402
C100	C3	Standard		100nF	0603
C101	A2	Power Section		1uF	0603
C102	B3	Power Section		1uF	0603
D2	B2	Standard	|	ESD3.3V	0402
D3	C2	Standard	|	ESD3.3V	0402
D4	C2	Standard	|	ESD3.3V	0402
D5	C2	Standard	|	ESD3.3V	0402
D6	C2	Standard	|	ESD3.3V	0402
D7	A2	Power Section	|	1N4148WS	D_SOD-323
D8	C2	Standard	|	LED Red/Yel	0805
D9	C2	Power Section	|	LED Red/Yel	0805
F2	C2	Standard		6V/200mA	0805
F3	B2	Standard		6V/200mA	0805
F4	A2	Standard		6V/200mA	0805
F5	A1	Standard		6V/200mA	0805
FL1	B1	Standard	|	TA0970A	SMD3838-6
FL2	C1	Standard	|	TA0970A	SMD3838-6
J1	B3	Standard	|	2x20 Socket	2x20 P2.54
J2	A2	Standard	|	U.FL GNSS Ext	U.FL
J4	C2	Standard	|	USB-C	USB-C 16P
J5	C2	Standard	|	U.FL 978	U.FL
J6	B1	Standard	|	U.FL 1090 Ext	U.FL
J7	B1	Standard	|	U.FL IFA	U.FL
J8	A1	Standard	|	U.FL GNSS Int	U.FL
J9※	C3	Power Section	|	MX1.25 Batt (Bottom)	MX1.25-2P
J10	C2	Standard	|	U.FL Airband	U.FL
J11※	A2	No Part	✕|	Debug pads 2… (Bottom)	2x10 bare pad
L1	B1	Standard		18nH 0402CS-…	0402
L2	A2	Standard		33nH	0402
L7	B3	Standard		6.8uH	0805
L8	C2	Standard		100nH	0402
L9	B3	Standard		7.5nH	0402
L10	B3	Standard		27nH	0402
L11	C3	Standard		6.8nH	0402
L12	C3	Standard		6.8nH	0402
L13	B3	Standard		7.5nH	0402
L14	B1	Standard		100nH	0402
L15	A1	Standard		33nH	0402
L16	C2	Power Section		1uH	SRN4018
L17	C3	Power Section		4.7uH XEL403…	XxL4030
L18	C3	Standard		100nH	0402
Q2	C3	Standard	|	AO3401A	SOT-23
Q3	B2	Standard	|	AO3401A	SOT-23
Q4	A2	Standard	|	AO3401A	SOT-23
Q5	A1	Standard	|	AO3401A	SOT-23
R1	A2	Standard		10k	0603
R2	A3	Standard		4.7k	0603
R3	A3	Standard		4.7k	0603
R4	A1	Standard		1k	0603
R5	A2	Standard		10k	0603
R6	B2	Standard		10k	0603
R7※	C1	No-Power Variant	✕	5.1k DNP	0603
R8※	C1	No-Power Variant	✕	5.1k DNP	0603
R9	B2	Standard		27R	0603
R10	B2	Standard		27R	0603
R11	B1	Standard		3.32k	0402
R17	C3	Standard		10k	0603
R18	B2	Standard		10k	0603
R19	B2	Standard		52.3R	0402
R21※	B2	Standard		0R	0603
R22※	B1	Standard		0R	0603
R23※	B2	Standard		0R	0603
R24※	B1	Antenna Select	✕	0R	0402
R25※	B1	Antenna Select	✕	0R	0402
R26	A2	Standard		10k	0603
R27	A2	Standard		10k	0603
R30	B2	DNP	✕	1k DNP	0402
R31	B2	Standard		1k	0603
R32	B2	Standard		10k	0603
R33	B2	Standard		100k	0603
R34	A2	Standard		10k	0603
R35	A2	Standard		10k	0603
R36	B2	DNP	✕	1k DNP	0603
R37	C1	Power Section		0R	0603
R38	C2	Power Section		180R	0603
R39	C2	Power Section		5.62k	0603
R40	C2	Power Section		31.6k	0603
R41	C3	Power Section		470k	0603
R42	C3	Power Section		150k	0603
R43	C1	Power Section		1k	0603
R44	C1	Power Section		10k	0603
R45	C2	Power Section		10k	0603
R46	C2	Power Section		10k	0603
R47	B3	Standard		10k	0603
R48	C1	Standard		0R	0603
R49	C1	Standard		0R	0603
R50	B2	Standard		30k	0603
R51	B2	Standard		10k	0603
R52	A2	Standard		10k	0402
R53	A2	Standard		10k	0402
R54	B2	Standard		10R	0402
R55	A2	Standard		10k	0603
R56	B2	Standard		10k	0603
R57	B2	Standard		33R	0603
R58	C3	Standard		10k	0603
R59	C3	Standard		10k	0603
R60	C3	Standard		0R	0603
R61	C3	Standard		10k	0402
R63	A3	Power Section		2.2k	0603
R64	C2	Standard		1k	0603
R65	C2	Power Section		1k	0603
R66※	C3	Debug Position	✕	0R	0603
RT1※	A3	Power Section		10k NTC (Bottom)	0603
SW3	C2	Power Section	|	TS24CA	TS24CA side
U1	A3	Standard	|	ME6211C33	SOT-23-5
U2	A3	Standard	|	TPS7A2033	SOT-23-5
U3	B3	Standard	|	TPS7A2033	SOT-23-5
U4	A2	Standard	|	BNO085	LGA-28
U5	A2	Standard	|	BMP388	LGA-10
U6	A3	Standard	|	QMC5883P	LGA-16
U7	A2	Standard	|	ATGM336H	LCC-18
U8	B2	Standard	|	RP2040	QFN-56
U9	B2	Standard	|	W25Q128	SOIC-8
U10	B3	Standard	|	CC1312R	QFN-48
U11	B1	Standard	|	QPL9547	DFN-8
U12	C1	Standard	|	BGA2817	SOT-363
U14	B2	Standard	|	AD8313	MSOP-8
U15	B2	Standard	|	TLV3501	SOT-23-6
U16※	B1	Antenna Select	|	XA17-G4K	SOT-363
U17※	A2	Standard	|	XA17-G4K	SOT-363
U18※	C1	Power Section	|	CH224K	ESSOP-10
U19	C2	Power Section	|	SY6970	QFN-24
U20	C3	Power Section	|	SY7069	TSOT-23-6
U21	C3	Standard	|	BK4819QN32SC	QFN-32
U22	C3	Standard	|	PCM1808PWR	TSSOP-14
U23	A3	Power Section	|	TPS7A2033	SOT-23-5
Y1	B1	Standard	|	12MHz CL=10p…	3225-4
Y2	B3	Standard	|	48MHz	3225-4
Y3	B2	Standard	|	32.768k	3215-2
Y4	C3	Standard	|	26MHz TCXO <…	2520-4
ZP1※	A1	IFA Match	✕	DNP	0603
ZP2※	B1	IFA Match	✕	DNP	0603
ZS1※	A1	IFA Match		0R	0603

229 designators, sorted by reference | Standard 178 · Power Section 37 · No-Power Variant 2 · Antenna Select 3 · IFA Match 3 · Debug Position 2 · DNP 2 · No Part 2 | 51 are polarised

Pilot Kit Expansion Board V4.10 Selective Placement Notes

Standard parts are fitted unconditionally and are not listed here. Every class below carries a precondition; getting it wrong breaks the function or the board — full reasoning in SELECTIVE_PLACEMENT-zh_CN.md.

No-Power Variant (2 parts, default here: DNP)

5.1k CC pull-downs, fitted on the no-power variant only. Together with U18 (CH224K) they skew the CC resistance, so PD negotiation may request 9V/12V into a board designed for 5V. Both sit inside a NO-PWR VER silkscreen box.

- R7 5.1k DNP [DNP] Not fitted on this (powered) variant
- R8 5.1k DNP [DNP] Not fitted on this (powered) variant

DNP (2 parts, default here: DNP)

Marked DNP on the PCB — kept empty by design. Populating it would create a DC path or a driver conflict. Not a per-batch choice; never fill it in.

Others in this class: R30, R36

Antenna Select (3 parts, default here: See notes)

Three mutually exclusive ways to route the 1090 antenna path — fit at most one. R24+R25 together short the external antenna to the on-board IFA; U16 plus either bypass shorts a switch port.

- R24 0R [DNP] 0R bypass → hard-wires the external antenna; not fitted here
- R25 0R [DNP] 0R bypass → hard-wires the on-board IFA; not fitted here
- U16 XA17-G4K [Fit] RF switch (default option) — this is the one to fit

IFA Match (3 parts, default here: See notes)

Pi matching network for the on-board IFA. The three positions combine freely into an L or pi network and are not exclusive. The default ZS1=0R is a straight-through link, not a tuned match — final values come from a VNA sweep with the enclosure fitted.

- ZP1 DNP [DNP] Shunt, antenna side; not fitted by default
- ZP2 DNP [DNP] Shunt, radio side; not fitted by default. First sweep starts at 3.3pF
- ZS1 0R [Fit] Series position, default 0R link; start the first sweep at 3.6nH

Debug Position (2 parts, default here: Fit)

Normally fitted; when sweeping the IFA just keep the downstream path open — C53 may stay.

- C53 100pF [Fit] Sweeping the IFA through J7 only requires the path after C53 to be open; C53 may stay fitted when the downstream chain is unpopulated. Remove or isolate it only when the downstream chain is populated
- R66 0R [DNP] **Not fitted by default.** It isolates the 26 MHz probe point on J11 pin 1 from the BK4819 oscillator node: that branch runs 120 mm on In2 and would add ~17 pF to a TCXO whose load budget is only 10 pF ±10%. Hand-solder a 0 R link only when you want to probe the 26 MHz reference

Power Section (37 parts, default here: Fit)

Fitted on the powered variant only. On the no-power variant this whole group is omitted and R7/R8 are fitted instead.

- J9 MX1.25 Batt [Fit] Battery connector, on the bottom side
- RT1 10k NTC [Fit] NTC for battery temperature sensing, on the bottom side
- U18 CH224K [Fit] CH224K USB-C PD sink controller; it provides the CC pull-down internally

Others in this class: BT1, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C101, C102, D7, D9, L16, L17, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R63, R65, SW3, U19, U20, U23

No Part (2 parts, default here: —)

Bare copper, a solder-bridge jumper or a bare test-pad array — there is no part to fit.

- ANT1 IFA Antenna [DNP] The on-board IFA — it is the PCB copper itself
- J11 Debug pads 2… [DNP] 2x10 bare pad array (bottom side): nothing to fit, mates with a 2x10p test clip. cJTAG + SWD + BOOTSEL/reset/self-test loopback. Bridge with tweezers or solder: 5-6 = reset, 7-8 = BOOTSEL, 19-20 = self-test (remove R57 first)

Standard (178 parts, default here: Fit)

This whole class is fitted unconditionally; only the few with an extra caveat are listed. The remaining 174 just get fitted.

- R21 0R [Fit] 0R link; open it to probe the AD8313 detector output on its own
- R22 0R [Fit] 0R link; open it to drive the 1090 switch A pin by hand
- R23 0R [Fit] 0R link; open it to drive the 1090 switch B pin by hand
- U17 XA17-G4K [Fit] GNSS-side RF switch. There is no bypass resistor on this side, so leaving it out means no signal path at all